A Global Dissipativity and Equilibrium-Stability Atlas for Lorenz–63

HCS Research Program

29 August 2026 (revision 1)

Abstract

We close an all-parameter theorem for the Lorenz–63 equations rather than infer a parameter continuum from the familiar numerical attractor. For every $\sigma, \beta > 0$ and $\rho \in \mathbb{R}$, a shifted quadratic form gives an exact dissipation ledger, a global forward bound, and an explicit absorbing ellipsoid. We list every equilibrium, factor the origin spectrum, and reduce the symmetric wings' cubic Routh–Hurwitz condition to one affine margin. This yields the exact stability split and, when $\sigma > \beta + 1$, the surface $\rho_H = \sigma(\sigma + \beta + 3)/(\sigma - \beta - 1)$ with its imaginary frequency. Zero σ, zero β, their intersection, and the $\rho = 1$ merger are resolved as separate singular faces. Only equilibrium and linear stability are claimed: neither nonlinear Hopf criticality nor universal chaos above ρ_H is inferred. Independent exact reconstruction, symbolic checks, canonical replay, and hostile mutations audit the result.

1 Convention and global dissipativity

We freeze the Lorenz–63 system

$$\dot{x} = \sigma(y - x), \qquad \dot{y} = x(\rho - z) - y, \qquad \dot{z} = xy - \beta z, \tag{1}$$

on $\mathbb{R}^3$, with $\sigma, \beta > 0$, $\rho \in \mathbb{R}$, and physical time t. It is equivariant under $(x, y, z) \mapsto (-x, -y, z)$, and its divergence is $-(\sigma + \beta + 1)$. Volume contraction alone will not be used as a dissipativity proof.

Theorem 1 (Explicit absorbing ellipsoid)**.** *Let $c = \rho + \sigma$,*

$$V = x^2 + y^2 + (z - c)^2, \qquad \kappa = \min\{2\sigma, 2, \beta\}.$$

Every solution of (1) is global forward and

$$\dot{V} = -2\sigma x^2 - 2y^2 - \beta z^2 - \beta(z - c)^2 + \beta c^2 \tag{2}$$

$$\le -\kappa V + \beta c^2, \tag{3}$$

$$V(t) \le e^{-\kappa t} V(0) + \frac{\beta c^2}{\kappa}(1 - e^{-\kappa t}). \tag{4}$$

Hence every ellipsoid $V \le R$ with $R > \beta c^2/\kappa$ absorbs every trajectory.

Proof. Differentiate V along (1). The xyz terms cancel, while the remaining xy coefficient vanishes because $c = \rho + \sigma$. The last term is closed by

$$-2\beta z(z - c) = -\beta z^2 - \beta(z - c)^2 + \beta c^2,$$

which proves (2) and (3). Gronwall's inequality gives (4). The vector field is polynomial, so a maximal finite-time solution could fail only by escaping every bounded set; (4) excludes that. On $V = R > \beta c^2/\kappa$, (3) points strictly inward, and (4) gives eventual entry. □

2 Equilibria and exact characteristic polynomials

Theorem 2 (Complete positive-parameter atlas)**.** *The origin O exists for all ρ and has*

$$\chi_O(\ell) = (\ell + \beta)[\ell^2 + (\sigma + 1)\ell + \sigma(1 - \rho)]. \tag{5}$$

It is asymptotically stable for $\rho < 1$, nonhyperbolic at $\rho = 1$, and a saddle for $\rho > 1$. If and only if $\rho > 1$, there are two further equilibria

$$E_\pm = (\pm\sqrt{\beta(\rho - 1)}, \pm\sqrt{\beta(\rho - 1)}, \rho - 1), \tag{6}$$

with common polynomial

$$\chi_E(\ell) = \ell^3 + (\sigma + \beta + 1)\ell^2 + \beta(\sigma + \rho)\ell + 2\sigma\beta(\rho - 1). \tag{7}$$

Proof. The equilibrium equations give $y = x$, $xy = \beta z$, and $x(\rho - z - 1) = 0$, proving the list. Substitution in

$$Df = \begin{pmatrix} -\sigma & \sigma & 0 \\ \rho - z & -1 & -x \\ y & x & -\beta \end{pmatrix}$$

gives (5) and (7). The quadratic Routh criterion in (5) gives the origin classification. □

3 The wing stability surface

For the cubic (7), all coefficients are positive when $\rho > 1$. The remaining Routh–Hurwitz condition is

$$\begin{aligned} D &= (\sigma + \beta + 1)(\sigma + \rho) - 2\sigma(\rho - 1) \\ &= \sigma(\sigma + \beta + 3) + (\beta + 1 - \sigma)\rho. \end{aligned} \tag{8}$$

Proposition 1 (One crossing or none)**.** *If $\sigma \le \beta + 1$, both wings are asymptotically stable for every $\rho > 1$. If $\sigma > \beta + 1$, define*

$$\rho_H = \frac{\sigma(\sigma + \beta + 3)}{\sigma - \beta - 1}. \tag{9}$$

Then $\rho_H > 1$; the wings are stable for $1 < \rho < \rho_H$, lie on a linear Hopf boundary at equality, and are unstable for $\rho > \rho_H$. At equality,

$$\chi_E(\ell) = (\ell + \sigma + \beta + 1)[\ell^2 + \beta(\sigma + \rho_H)], \tag{10}$$

so the crossing pair is $\ell = \pm i\sqrt{\beta(\sigma + \rho_H)}$.

Proof. Equation (8) is positive for all $\rho > 1$ when its slope is nonnegative. For negative slope it has the unique zero (9), and

$$(\rho_H - 1)(\sigma - \beta - 1) = (\sigma + 1)(\sigma + \beta + 1) > 0.$$

The cubic Routh–Hurwitz criterion gives the sign classification. Substituting (9) into (7) gives (10). This is a linear spectral crossing; its nonlinear criticality is not part of the proposition. □

4 Singular parameter faces

The strict positivity assumption is structural. If $\sigma = 0$ and $\beta > 0$, $x = s$ is constant and the equilibria form

$$E_s = \left(s, \frac{\beta\rho s}{\beta + s^2}, \frac{\rho s^2}{\beta + s^2}\right), \qquad \chi_s = \ell[\ell^2 + (1 + \beta)\ell + \beta + s^2]. \tag{11}$$

If $\beta = 0$ and $\sigma > 0$, the equilibria are $(0, 0, z_0)$ and

$$\chi_{z_0} = \ell[\ell^2 + (\sigma + 1)\ell + \sigma(1 - \rho + z_0)]. \tag{12}$$

They are transversely stable for $z_0 > \rho - 1$, saddles for $z_0 < \rho - 1$, and nonhyperbolic at equality. At $\sigma = \beta = 0$ the equilibrium set is the union of $(0, 0, z)$ and $(x, 0, \rho)$. The tangent zero roots make clear why these faces are not silently included in Theorem 1. At $\rho = 1$, (6) merges with the origin; we claim the resulting nonhyperbolic boundary, not a center-manifold normal form.

5 Source and claim boundary

Lorenz [1] introduced (1). Pade, Rauh and Tsarouhas [2] treat nonlinear Hopf direction, which is outside our theorem. Tucker [3] and Guckenheimer–Williams [4] give rigorous classical-parameter and geometric-attractor context. We do not extrapolate those results to a global chaos classification: above ρ_H the proved statement is linear instability of $E_\pm$, no more.

References

[1] E. N. Lorenz, *Deterministic Nonperiodic Flow*, Journal of the Atmospheric Sciences 20 (1963), 130–141. DOI link.

[2] J. Pade, A. Rauh and G. Tsarouhas, *Analytical investigation of the Hopf bifurcation in the Lorenz model*, Physics Letters A 115 (1986), 93–96. DOI: 10.1016/0375-9601(86)90031-9.

[3] W. Tucker, *A Rigorous ODE Solver and Smale's 14th Problem*, Foundations of Computational Mathematics 2 (2002), 53–117. DOI: 10.1007/s002080010018.

[4] J. Guckenheimer and R. F. Williams, *Structural stability of Lorenz attractors*, Publications Mathématiques de l'IHÉS 50 (1979), 59–72. DOI: 10.1007/BF02684769.

Declarations

Scope. `NO_BAD_EULER_OR_ROOT_NUMBER`; no target arithmetic, target-zero, global-chaos, or Hilbert–Pólya claim. **Data and code.** The exact ledger and all audit programs accompany HCS-C227. **AI-use disclosure.** Generative tools assisted drafting and code generation; the released internal checks validate the displayed identities and metadata. This is not external peer review.